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Using ontology-based semantic similarity to facilitate the article screening process for systematic reviews

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ABSTRACT

Objective: Systematic Reviews (SRs) are utilized to summarize evidence from high quality studies and are considered the preferred source of evidence-based practice (EBP). However, conducting SRs can be time and labor intensive due to the high cost of article screening. In previous studies, we demonstrated utilizing established (lexical) article relationships to facilitate the identification of relevant articles in an efficient and effective manner. Here we propose to enhance article relationships with background semantic knowledge derived from Unified Medical Language System (UMLS) concepts and ontologies.

Methods: We developed a pipelined semantic concepts representation process to represent articles from an SR into an optimized and enriched semantic space of UMLS concepts. Throughout the process, we leveraged concepts and concept relations encoded in biomedical ontologies (SNOMED-CT and MeSH) within the UMLS framework to prompt concept features of each article. Article relationships (similarities) were established and represented as a semantic article network, which was readily applied to assist with the article screening process. We incorporated the concept of active learning to simulate an interactive article recommendation process, and evaluated the performance on 15 completed SRs. We used work saved over sampling at 95% recall (WSS95) as the performance measure.

Results: We compared the WSS95 performance of our ontology-based semantic approach to existing lexical feature approaches and corpus-based semantic approaches, and found that we had better WSS95 in most SRs. We also had the highest average WSS95 of 43.81% and the highest total WSS95 of 657.18%.

Conclusion: We demonstrated using ontology-based semantics to facilitate the identification of relevant articles for SRs. Effective concepts and concept relations derived from UMLS ontologies can be utilized to establish article semantic relationships. Our approach provided a promising performance and can easily apply to any SR topics in the biomedical domain with generalizability.

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1. Introduction

Evidence-based practice (EBP) relies on up-to-date research evidence to inform clinical decisions. By 2020, the Institute of Medicine (IOM) seeks to ensure that 90% of clinical decisions are evidence-based [1]. Systematic reviews (SRs), or systematic reviews of literature, are utilized to summarize evidence drawn from high quality studies and are considered the preferred source of EBP [2].

SRs are commonly conducted by domain experts who are able to identify the SR scopes and research questions, generate search strategies to retrieve citations from related databases (e.g. MEDLINE), assess study quality, and synthesize evidence [3,4]. However, conducting an SR can be labor intensive and time consuming [5], generally requiring several months to complete [6]. Specifically, a typical exhaustive search would yield search results ranging from 300 to more than 10,000 citations [6,7]. Before synthesizing relevant evidence, expert researchers need to classify citations (relevance = inclusion, irrelevance = exclusion) based on their *title* and *abstract*. Then through the triage process, included articles will proceed to the full-text level. In most SRs, expert researchers include only 2–30% of citations at the title and abstract level triage, and 1.6–27% are included at the full-text level [6]. In other words, expert researchers spend most of their effort excluding irrelevant or low quality studies. As the article screening process (classification of articles) is one of the most resource and time intensive steps, such workload and resource challenges can limit the tractability of an individual review, the ability to fund a

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review, and also the ability to respond to new evidence that may require an update to an existing review. There has been criticism of outdated evidence in published SRs because of the inefficient production of SRs [4,8,9].

Existing studies have shown that automated citation classifications are a valuable tool to accelerate the article screening process and save workloads in SRs, as they utilize machine learning approaches (i.e. Naïve Bayes and Support Vector Machine) [6,10–15] to classify and rank documents. With carefully-designed training processes, they have demonstrated the ability to yield good performance in recall, precision, and/or workload savings. However, supervised machine learning approaches rely on the availability of expensive manual annotations, and the learned models have limited generalizability to new SR topics. Besides, some systems are not immediately accessible for non-computer science specialists. To promote the usability in practical scenarios, (semi-supervised) active learning has been introduced to make effective use of human efforts. It interactively queries reviewers for their manually classified citations as training datasets, and then triggers an automated classification process [11,13]. Some enhancements on active learning like weighting, detection of proper training size, and covariate shift, further close the gaps between SR requirements and technical solutions [15]. However, these machine learning approaches are still not widely used due to the major obstacle of significant limited generalizability [16].

In our previous study [17], we demonstrated using established and pre-determined article relationships, and incorporated the concept of active learning to iteratively recommend articles and receive relevance feedback (RF) from human SR researchers. Unlike the traditional machine learning methods, we generated an article network representing the relationship (similarity) among articles. Our article network was built based on the lexical similarity of any paired articles, using several MEDLINE elements [18] from an article, including *title*, *abstract*, *MeSH*, *author*, and *publication type*. The article network was readily applied to assist with the article screening process where articles most likely to be relevant were iteratively recommended based on pre-established article relationships in the article network and RF. Without relying on prior supervised training data, we obtained competitive workload savings compared to supervised learning approaches. Besides, the active refinement of our recommendation model using RF is almost in real-time. In addition, we visualized the established article network with a force-directed layout to examine the distribution of articles and hidden structural patterns [19]. Our results revealed that relevant (included) articles tended to aggregate together in some module classes, providing further evidence of the existence of strong relationships among included articles. In both of our previous studies, we have demonstrated utilizing article relationships to accelerate the article screening process for SRs.

To further improve the precision of article relationships, in this study we implemented semantic features to advance the article similarity calculation. For instance, *titles* or *abstracts* may contain different terminologies (e.g. "dry mouth" and "xerostomia") that have similar topics or concepts (e.g. oral symptom), which is beyond the capacity of exact matching or lexical features. Existing studies have utilized distributional semantics, which is also referred to as the word embedding [20] (or Brown clustering [21]) approach using word co-occurrences in large corpus contexts, to capture semantically similar words and represent words with semantic vectors [14,22]. With this notion, related words (or terms) are expected to have similar vectors, and text pieces can be represented as weighted combinations of word vectors. The word embedding approach requires an unsupervised or semi-supervised learning process and relies on the availability of a large and high-quality corpus. While there are generic models which have been trained for direct uses [20,23], their accuracy in reaching concept or literature semantic relationships in the biomedical domain remains uncertain.

A work by Meystre (2008) conducted a thorough review on natural language processing (NLP) approaches applied to the clinical domain and summarized that standard ontologies and terminologies could be utilized as background knowledge for information extraction and semantic analysis [24]. Unified Medical Language System (UMLS) [25], integrates, classifies, and distributes key terminologies in biomedical informatics, and has been a well-recognized resource for knowledge processing, maintenance, translation, and other utilizations. Existing studies have applied the UMLS concept semantic similarity which is derived from concept relations in the biomedical ontologies within the UMLS framework (UMLS contained ontologies, or UMLS ontologies) [26–30]. Ontology-based approaches, such as intrinsic information content (IC), have shown to provide better performance than corpus-based approaches, and compute similarity values highly correlated with human ratings in the biomedical domain [26,31].

Motivated by the capability of UMLS ontologies in assembling biomedical knowledge, we proposed to augment article relationships with semantic similarities derived from UMLS concepts. In this paper, we built a pipelined semantic concepts representation process to represent articles from an SR into a semantic space of UMLS concepts, and represented article relationships as a semantic article network. We mapped articles into a concepts representation (set of concepts) through three steps: (1) concept annotation, (2) concept optimization, and (3) concept expansion. Throughout the process, we leveraged concept relations maintained in UMLS ontologies (SNOMED-CT [32] and MeSH [33]) and utilized ontology-based concept similarities to prompt concept features in each article's profile. At the end of the three-step process, articles were represented in a weighted semantic vector space. Augmented article relationships were readily established as a semantic article network, which can be applied to the article screening process in an efficient manner. Consider a scenario where article content does not explicitly carry certain terms or concepts, it would be difficult to find such articles using only the lexical relationships or exact concept matching. Our approach could improve the identification of these articles in SRs. In this paper, we also compared our semantic approach with studies using lexical features, Cohen [10], Matwin [13], and Ji [17] as well as studies using corpus-based semantic approaches, Khabsa [14] and Jonnalagadda [22].

2. Methods

2.1. Data source

We used 15 completed SRs produced by the Drug Effectiveness Review Project team (DERP) for performance evaluation [34]. The datasets were made publicly available from Cohen 2006 [6], and were utilized by many related studies, i.e., Cohen [10], Matwin [13], Jonnalagadda [22] and Khabsa [14]. These 15 SRs were completed by experienced and knowledgeable human SR researchers, with inclusion and exclusion decisions made by at least two expert reviewers. Table 1 shows the number and percentage of articles included at the (1) abstract level decision and the (2) full-text level decision. Using the PubMed Identifier (PMID), we downloaded the full references in MEDLINE format and extracted article elements. In our previous study [17], we built an article network representing the relationship (similarity) among articles based on the lexical similarity of any paired articles using several MEDLINE elements [18], including *title*, *abstract*, *MeSH*, *author*, and *publication type*. In this study, we implemented semantic features to advance the article similarity calculation for *title* and *abstract*.

Table 1
DERP 15 SRs' total article numbers and rate of inclusion.

	Total	Abstract N (%)		Full-text N (%)	
ACE Inhibitors	2544	183	(7.19%)	41	(1.61%)
ADHD	851	84	(9.87%)	20	(2.35%)
Antihistamines	310	92	(29.68%)	16	(5.16%)
Atypical Antipsychotics	1120	363	(32.41%)	146	(13.04%)
Beta Blockers	2072	302	(14.58%)	42	(2.03%)
Calcium Channel Blockers	1218	279	(22.91%)	100	(8.21%)
Estrogens	368	80	(21.74%)	80	(21.74%)
NSAIDS	393	88	(22.39%)	41	(10.43%)
Opioids	1915	48	(2.51%)	15	(0.78%)
Oral Hypoglycemics	503	139	(27.63%)	136	(27.04%)
Proton Pump Inhibitors	1333	238	(17.85%)	51	(3.83%)
Skeletal Muscle Relaxants	1643	34	(2.07%)	9	(0.55%)
Statins	3465	173	(4.99%)	85	(2.45%)
Triptans	671	218	(32.49%)	24	(3.58%)
Urinary Incontinence	327	78	(23.85%)	40	(12.23%)

In the following sections, we describe (1) the ontologies we used from the UMLS framework, (2) the semantic concepts representation development process for each *title* and *abstract*, (3) the development of the semantic article network, and (4) the evaluation setting.

2.2. UMLS knowledge repository

UMLS Concepts: UMLS [25] is a compendium of many controlled vocabularies in the biomedical domain, as well as a comprehensive thesaurus of biomedical concepts. The UMLS Metathesaurus consists of over two million concepts with a mapping structure from nearly 200 different vocabularies (ontologies). Specifically, it is organized by concepts (senses or meanings) associated with attributes and names in source vocabularies, and is enumerated via Concept Unique Identifiers (CUIs). The Metathesaurus also preserves useful relationships between concepts from each vocabulary. We used MetaMap [35] to retrieve UMLS concepts from biomedical text. MetaMap assigns CUIs to words and terms using rules and patterns. Thereby, we can overcome the barrier of when the same concepts are expressed in a variety of ways in different contexts or by different authors.

Ontologies: Systematized Nomenclature of Medicine-Clinical Terms (SNOMED-CT) [32] and Medical Subject Headings (MeSH) [33] are two of the most widely used knowledge sources within the UMLS framework. SNOMED-CT has been the main ontology used in the biomedical domain [31]. It provides a global and broad hierarchical terminology that models clinical data (i.e. storage, encoding, and retrieval) in healthcare. Specifically, the US edition of SNOMED-CT covers more than 300,000 unique concepts classified in 19 top-level hierarchies. MeSH is a controlled vocabulary for indexing published studies from the MEDLINE database. It provides over 25,000 biomedical concepts hierarchically classified in 16 categories. MeSH has been widely used for information retrieval, but can also be used for other purposes, such as encoding and storing biomedical data [36–41]. Although MeSH is not explicitly claimed as an ontology, it provides broader-narrower relations in its hierarchy [42]. Several studies have utilized MeSH as an ontology for semantic analysis and achieved high correlations with human ratings; some studies also applied cross-ontology or multi-ontology approaches by using MeSH together with SNOMED-CT and obtained improved performance over using SNOMED-CT alone [31,36–41]. Inspired by the properties of MeSH and prior studies, in our study, we used the vocabulary concepts and the hierarchical (hyponym/hypernym) relations in both SNOMED-CT and MeSH as our ontology knowledge sources. By loading the Rich Release Format (RRF) [43] set of UMLS Metathesaurus into MySQL database, we were able to access vocabularies and concept relations in SNOMED-CT and MeSH from relational tables (i.e. MRREL and MRHIER) [43]. Using SNOMED-CT and MeSH together enables a broader concept and knowledge coverage. In this study, we used the UMLS Metathesaurus Insertion Version 2015AB.

2.3. Semantic concepts representation development process

A proper representation of articles is an important prerequisite for establishing accurate semantic relationships among articles. We created a three-step process (Fig. 1) to develop concepts representation for each *title* and *abstract*. The three-step process includes (1) concept annotation, (2) concept optimization, and (3) concept expansion. The output of this process is an optimized and enriched concepts representation, or a semantic vector space, which can be used to establish article relationships as a semantic article network.

Concept Annotation: We annotated all terms and words in the *title* and *abstract* from each article with MetaMap API [44], and mapped them to UMLS CUIs. When using MetaMap, we did not restrict the mapping score or semantic type in order to investigate a more generalizable approach. We then parsed the MetaMap output to create a list of concepts. In the results, each article can be represented as two separate lists of concepts, one as the *title* concept list and the other as the *abstract* concept list. At this stage, we included all CUIs that matched or unmatched to SNOMED-CT and MeSH. We called this the all-inclusive concepts representation. In addition, each concept contains three key components: CUI, preferred concept name (identified by UMLS), and frequency (N). The frequency (N) indicates the total number of times that a concept is mapped from words and phrases in the *title* or *abstract*.

After the concept annotation, we were able to establish an all-inclusive concepts representation for each article, which includes biomedical specific concepts, as well as other more general concepts. For instance, concept "*C188335: To [Qualitative Concept],*" derived from the trivial stop-word "*to*", can be seen as a noise; analogous concepts "*C1706408: PLACEBO [Research Activity]*", "*C0032042: Placebo [Therapeutic or Preventive Procedure]*", and "*C1696465: PLACEBO [Biomedical or Dental Material],*" derived from "*placebo*" in the same context, lead to overmatching; and concept "*C00449851: Methods [Techniques]*" is too general to provide meaningful semantic connections. The contribution of these concepts should depend on the requirements of a specific SR.

Concept Optimization: To optimize the concept list, a tunable process of concept categorization (Fig. 2) was implemented to control noises, overmatching concepts, and concepts with low specificity. We utilized SNOMED-CT and MeSH to identify concepts commonly used in the biomedical/clinical domain. Concepts not matched in SNOMED-CT and MeSH were considered as *unmatched*

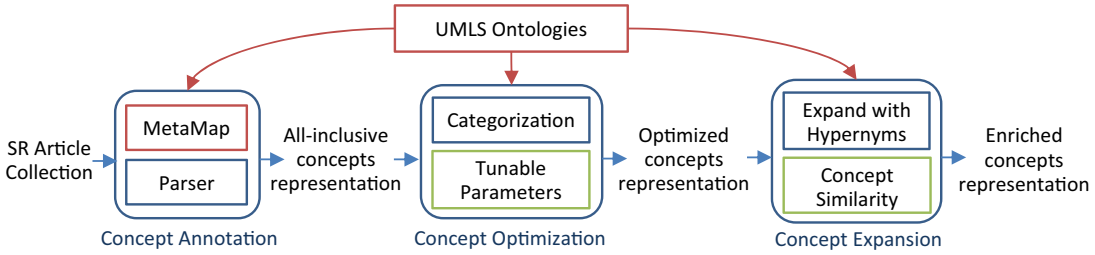


Fig. 1. Semantic concepts representation development process.

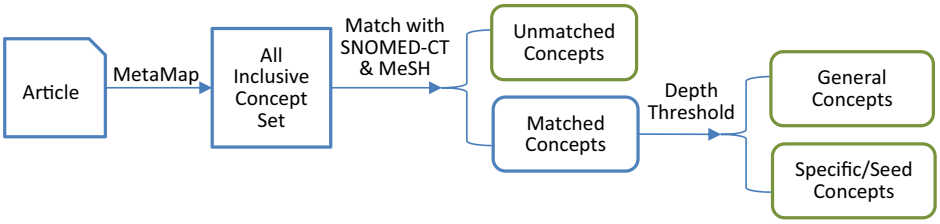


Fig. 2. Concept categorization process.

concepts, which were usually noise or overmatching as abovementioned. Concepts matched to SNOMED-CT and MeSH proceeded to the next stage to be categorized as general concepts or specific concepts. In an ontology, concepts are arranged in a hierarchical structure, which is also a directed acyclic graph with a root node (considered as a taxonomic structure, or a taxonomic tree). Concepts with lower depths, located closer to the root, have broader meanings; concepts with higher depths, located farther away from the root, are hyponyms with more specific meanings. We used a *depth threshold* to categorize concepts based on their depths in the taxonomic tree. In this sense, concepts with a depth lower than the depth threshold were categorized as *general concepts*; concepts with a depth equal to or higher than the threshold were categorized as *seed concepts*. Because of the poly-hierarchical structure in SNOMED-CT and MeSH, one concept can have multiple parents and depths. We calculated both its smallest and largest depth, but used the smallest depth for concept categorization and weighting. This helped avoid biases resulting from undesired lengthy paths assigned to some general concepts.

We also assigned weights to concepts and used parameters to control the contribution of each concept category. We assigned weights to *seed concepts* based on their frequency from the annotation stage and depth in the taxonomic tree ($weight_{seed} = frequency \cdot depth$). For *general concepts*, we had a ratio, α ($0 \leqslant \alpha \leqslant 1$), to control their contributions ($weight_{general} = \alpha \cdot frequency \cdot depth$). For *unmatched concepts*, we used a controlling ratio, β ($0 \leqslant \beta \leqslant 1$), ($weight_{unmatched} = \beta \cdot frequency$). While various SRs can have very different scopes, criteria, or preferences, there are no universal parameters that can be applied to all scenarios [17]. To leverage prior knowledge in SRs (i.e. researchers' experience), parameters should be flexibly adjustable.

Concept Expansion: The purpose of concept expansion is to overcome the limitation of exact concept matching, and provide an exhaustive concepts representation for each article [28,45,46]. Consider a scenario when two articles are semantically related but share no identical concepts in their concept sets. With exact concept matching, it would be difficult to find connections, thus missing potentially important semantic features. A concept expansion to include related concepts (i.e. hyponyms, hypernyms, and siblings) in the ontology may solve the problem. We leveraged concept relations maintained in the taxonomic structure of SNOMED-CT and MeSH. For each *seed concept*, we expanded it by including all ancestors (hypernyms) that have a higher (or equal) depth than the assigned depth threshold. To access concept relations in the taxonomic structure, we loaded an RRF [43] set of UMLS Metathesaurus into a MySQL database. We used C++ to index the path-to-root (PTR) from the table MRHIER, which contains concept relation information of the taxonomic structure.

Fig. 3 illustrates a concept expansion example. *C4 (Animal bite)* and *C6 (Stung by sea anemone)* are *seed concepts* representing in two different articles; both *C4* and *C6* are hyponyms of *C1 (accident due to animal or plant)* as shown in Fig. 3(a). If a SR scope is to analyze accidents due to environment factors, *C4* and *C6* should contribute to the similarity of the two articles. In Fig. 3(b), we expanded *C4* with *C1* and *C2*, and expanded *C6* with *C1* and *C3*. The two expanded concept sets now share a commonality, *C1*, which grants similarity between the two articles.

To control the contribution of *expanded concepts*, we assigned weights to them based on their semantic similarity to the *seed concepts*. Since the expansion only utilized concept relations between a hypernym and a hyponym, we deployed a simple version of an ontology-based method using the distance (shortest path) between concepts [31], to capture concept semantic similarity. In this sense, the weight of an *expanded concept* is calculated with the formulas below. If one *expanded concept* acts as a hypernym of multiple *seed concepts*, then the highest weight was used.

$$\text{Similarity}(C_{seed}, C_{expansion}) = \frac{1}{distance(C_{seed}, C_{expansion}) + 1}$$

$$Weight_{expansion} = Weight_{seed} \cdot \text{Similarity}(C_{seed}, C_{expansion})$$

At the end of this process, each *title* or *abstract* was represented with an optimized and enriched set of concepts; the entire article collection from an SR can then be represented in a weighted semantic vector space that comprises all derived concepts as semantic features. Fig. 4 shows an example of a specific title along with its optimized and enriched concept set. Due to the space limitation, we only list part of the concepts under each category. For each concept, we list its CUI, name, and weight based on the abovementioned formula(s).

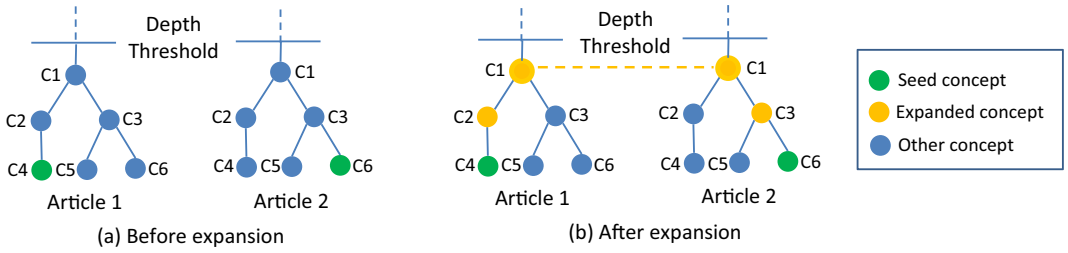


Fig. 3. Illustrated concept expansion.

PMID-12811500: Controlled, double blind, multicentre clinical trial to investigate long-term tolerability and efficacy of trospium chloride in patients with detrusor instability.

Specific/Seed Concepts
C0013072 Double-Blind Method *7*
C1096776 Multicenter Study *5*
C0077408 Trospium Chloride *6*
C1261562 Detrusor Instability *8*
Expanded Concepts
C0242896 Anticholinergic Agent *1.5*
C0005686 Urinary Bladder Disease *4*
General Concepts
C1292732 Investigate $\alpha \cdot 6$
C1280519 Efficacy $\alpha \cdot 4$
C0030705 Patients $\alpha \cdot 5$
Unmatched Concepts
C1883351 To $\beta \cdot 1$
C0443252 Long-term $\beta \cdot 1$
C3274448 Tolerability Study $\beta \cdot 1$

Fig. 4. Illustrated concept set.

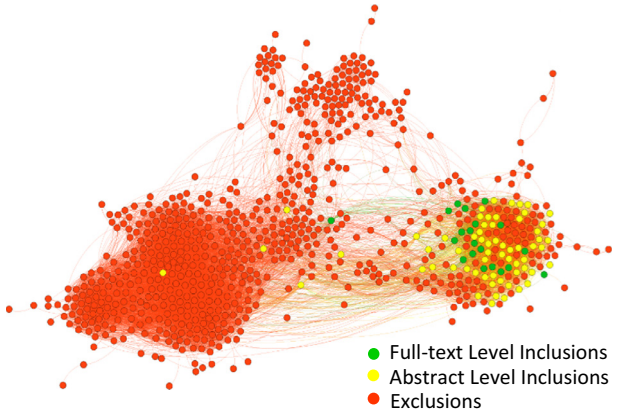


Fig. 5. Illustrated semantic article network.

2.4. Semantic article network

Semantic Similarity Score: With the semantic vector space established in the previous process, we were able to calculate semantic similarity scores for all article pairs using Cosine similarity. Cosine similarity is widely used in text mining, and it measures the cosine of the angle between a pair of vectors. It reflects the degree of similarity based on the presence and weights of concept features in each article's (*title* or *abstract*) concepts representation. The resulting similarity score ranges from 0 to 1, where 0 indicates independence and 1 means exactly the same. The final similarity score is a weighted sum of *title* similarity and *abstract* similarity, ranging from 0 to 2.

Article Network: We conceptualized relationships (similarities) between articles as a semantic article network. Under this concept, each article is modeled as a node in the network and the relationship between two articles is modeled as an edge connecting them. [17] Fig. 5 shows an illustrated network of a completed SR (*ADHD* [34]), which is displayed in a force-directed layout. Theoretically, the network should be almost a complete graph in which every pair of articles has an edge representing the similarity between them. For visualization purposes, we eliminated the edges with lower similarity scores to provide a more human readable network.

2.5. Evaluation setting

Simulated Interactive Recommendation Process: There is no human SR researcher in this process. As abovementioned, we used the 15 completed SRs from DERP to evaluate our approach. We established article relationships and created the semantic article network, and then incorporated the concept of active learning to iteratively recommend articles and receive relevance feedback (RF), which was simulated by inclusion and exclusion decisions documented in the completed SRs. We first recommended an article based on key questions and search strategies of the SR. Once a recommended article is classified as "included" or "excluded", an inclusion list (IN) and an exclusion list (EX) are created. We then iteratively recommended relevant articles based on the similarity to the IN. Assuming V is the set of all articles, and U (U = V-IN-EX) is the set of articles that have never been recommended, the recommendation can be formulated as:

$$\text{Recommendation} = \arg\max_{v \in U}\left(\sum_{x \in IN} \vec{vx} \cdot \vec{w}\right)$$

In the formula, $\vec{vx}$ is a two dimensional vector representing the similarity of *title* and *abstract* between article v and x. $\vec{w}$ is a weight parameter, which controls the contribution of *title* and *abstract*.

Measures: To evaluate our performance and compare it with others, we used work saved over sampling at 95% recall (WSS95). It is a performance measure first proposed by Cohen [6] to calculate the overall labor savings while maintaining the recall at 95%. This assumes that a recall of >0.95 is necessary for a document classification system. Precision should be as high as possible, as long as recall is at least 0.95. WSS95 has been commonly used for evaluating similar works [6,10,12–14,22], and it is calculated as below.

$$\text{WSS} = \frac{(TN+FN)}{N} - 1 + \frac{TP}{TP+FN}; \quad \text{WSS95} = \frac{(TN+FN)}{N} - 0.05$$

TP is the number of true positive (relevant) articles, TN is the number of true negatives (irrelevant) articles, FN is the number of false negative (relevant) articles, and N is the total number of articles in each report. We used paired *t*-test or repeated ANOVA with the Hommel multiple comparison adjustment to further explore statistical evidences. Hommel's method demonstrated

type I error protection with good power and is considered a better approach than Bonferroni or Holm [47,48]. We also plotted recall performance curves, in which the x-axis represents the percentage of articles screened, and the y-axis represents the recall rate. The recall performance curve provides the recall performance of the article recommendation process over time.

3. Results

3.1. Experiment setting

In this study, we had a restrictive experiment setting; we assigned a zero ratio ($\alpha, \beta = 0$) to *unmatched concepts* and *general concepts* to eliminate their contributions in the feature space. As no universal weight parameters can be identified in our preliminary study [17], we used an equal weight setting for *title* similarity and *abstract* similarity ($\vec{w} = [1, 1]$).

As for the depth threshold, with respect to the diversity of SR scopes and the different structures in SNOMED-CT and MeSH, we examined the WSS95 performance using SNOMED-CT (Fig. 6a) and MeSH (Fig. 6b) separately and jointly (Fig. 6c) with a series of depth thresholds (1–15). In Fig. 6, the 15 DERP SRs are represented in different colored performance curves. As shown in Fig. 6a and b, although SNOMED-CT and MeSH have different structures, they resulted in the same range of depth thresholds (1–9) with non-zero WSS95. We found that each individual SR has its own *best depth threshold* that led to the highest WSS95. SNOMED-CT had an average best depth threshold of 3.53; MeSH had an average best depth threshold of 4.87. When combining SNOMED-CT and MeSH together, the average best depth threshold was 3.53. As different depth thresholds resulted in varying WSS95 performance for individual SRs, and there is no universal best depth threshold that could be used for all SRs, a flexible depth threshold is needed. In this experiment, we concluded that we could apply the average best depth threshold (i.e. 3) or the depth threshold that resulted in higher performance in most SRs (i.e. 1) as the initial default setting for later adjustments.

3.2. Performance report

Table 2 presents the WSS95 performance with different settings: (1) SNOMED-CT, (2) MeSH, and (3) SNOMED-CT + MeSH. For each setting, we compared the performance resulting from a fixed depth threshold, 1 (DT = 1), for all SRs, and a flexible depth threshold, the best depth threshold, for each individual SR. As shown in Table 2, flexible depth thresholds brought improved performance over the fixed depth threshold with a WSS95 improvement of 2.40% on average. We then evaluated the WSS95 performance across different settings using flexible depth thresholds. Specifically, SNOMED-CT resulted in an average WSS95 of 41.15%; MeSH resulted in an average WSS95 of 36.97%. While SNOMED-CT brought better performance than MeSH in 10 SRs, MeSH outperformed SNOMED-CT in 5 SRs (e.g. Triptans). With SNOMED-CT + MeSH as the knowledge sources, we achieved an improved average WSS95 of 43.81%. Specifically, we obtained an improved performance in 10 SRs with SNOMED-CT + MeSH, compared to using SNOMED-CT alone. In summary, even though flexible depth thresholds resulted in the best performance, we can still receive competitive results using a fixed depth threshold. In addition, SNOMED-CT + MeSH can achieve better WSS95 compared to using SNOMED-CT or MeSH alone.

Fig. 7 presents the recall performance curves for all 15 SRs using SNOMED-CT + MeSH. We noticed that the speed (efficiency) of identifying relevant articles differs from one SR to another. For example, ACE Inhibitors and ADHD identified relevant articles rapidly at the early screening stage and achieved a high recall immediately; Atypical Antipsychotics and Urinary Inconvenience maintained a more stable recall speed for the entire process; Antihistamines had a speed acceleration in a much later stage. Overall, with a 40% article screening progress, 12 SRs can reach a recall of >70%.

3.3. Performance comparison to lexical features

We compared our semantic approach with prior works using lexical features, including (1) our lexical similarity article network approach, Ji and Yen [17], and (2) representative supervised machine learning approaches, Cohen [10] and Matwin [13], who applied lexical features in their learning and testing process. In these studies, the optimization process or intra-topic training process applied adjustable parameters to present the best performance in different SRs. To be comparable with these studies, we used the results obtained from the flexible depth thresholds. Table 3 shows the WSS95 comparison. We did not include Cohen 2006 [6] for side by side comparison as it is an earlier work of Cohen [10]. Cohen 2006 [6] reported an average WSS95 of 23.43%, using a machine learning approach of voting perceptron.

The repeated ANOVA test shows significant differences across four studies ($p = 0.035$), but the pairwise comparisons with the Hommel adjustment did not find significant differences ($p > 0.05$). However, when examining each individual SR, we outperformed Cohen [10] in 8 SRs; outperformed Matwin [13] in 13 SRs; and outperformed our lexical similarity approach, Ji and Yen [17] in 12 SRs. We also had the highest average WSS95 of 43.81% and the highest total WSS95 of 657.18%.

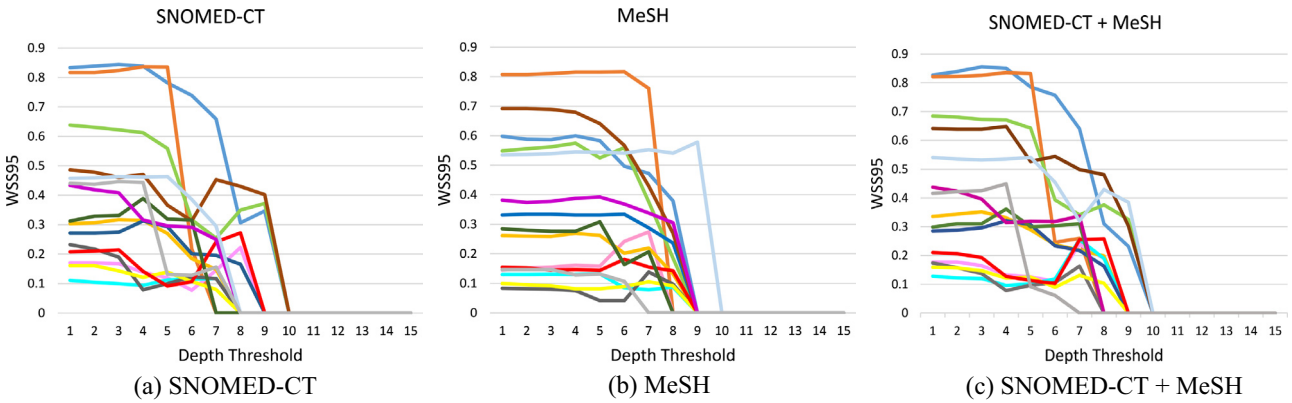


Fig. 6. Depth threshold and resulting WSS95 performance.

Table 2
Performance report on 15 DERP SRs with full-text level inclusion decisions.

SR Report Topic	SNOMED-CT		MeSH		SNOMED-CT + MeSH	
	WSS95% (DT = 1)	WSS95% (Flexible DT)	WSS95% (DT = 1)	WSS95% (Flexible DT)	WSS95% (DT = 1)	WSS95% (Flexible DT)
ACE Inhibitors	83.33	84.39 (3)	59.83	59.98 (4)	82.70	85.57 (3)
ADHD	81.67	83.67 (4)	80.73	81.67 (6)	82.14	83.55 (4)
Antihistamines	17.10	22.26 (8)	14.52	27.42 (7)	17.74	23.87 (7)
Atypical Antipsychotics	30.36	31.70 (3)	26.16	26.96 (4)	33.57	35.18 (3)
Beta Blockers	63.85	63.85 (1)	54.87	57.48 (4)	68.48	68.48 (1)
Calcium Channel Blockers	11.08	11.74 (7)	12.97	13.22 (5)	12.81	25.53 (7)
Estrogens	27.17	31.25 (4)	33.15	33.42 (2)	28.53	32.07 (4)
NSAIDS	48.60	48.60 (1)	69.21	69.21 (1)	64.12	64.89 (4)
Opioids	23.24	23.24 (1)	8.30	13.84 (7)	17.39	17.39 (1)
Oral Hypoglycemic	16.10	16.10 (1)	9.94	10.54 (7)	15.90	15.90 (1)
Proton Pump Inhibitors	20.78	27.23 (8)	15.45	18.15 (6)	21.01	25.81 (8)
Skeletal Muscle Relaxants	31.22	38.89 (4)	28.48	30.86 (5)	29.88	36.15 (4)
Statins	43.30	43.30 (1)	38.16	39.23 (5)	43.74	43.74 (1)
Triptans	45.75	46.35 (3)	53.50	57.82 (9)	54.10	54.10 (1)
Urinary Incontinence	44.04	44.65 (4)	14.68	14.68 (1)	41.59	44.95 (4)
Average	**39.17**	**41.15**	**34.66**	**36.97**	**40.91**	**43.81**

DT: Depth Threshold.

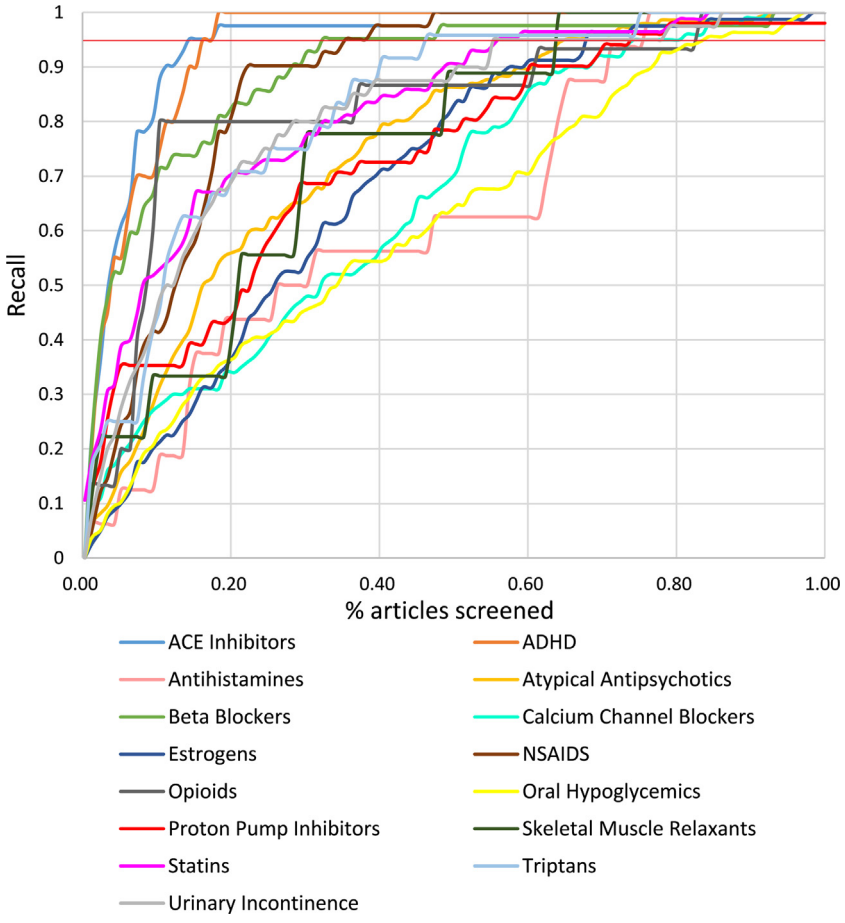


Fig. 7. Recall performance curves.

3.4. Performance comparison with corpus-based semantic approaches

With the same goal to facilitate the article screening process, Jonnalagadda [22] and Khabsa [14] used corpus-based semantic approaches with word embedding or Brown clustering. With the same DERP dataset, Jonnalagadda [22] reported WSS95 using the abstract level decision, and Khabsa [14] reported WSS at a various recall level using the full-text level decision. We examined performance at both abstract and full-text levels to have a side-by-side comparison with them. In addition, because Khabsa [14] reported WSS with various recall levels, ranging from 75.00% to 99.50%, for each individual SR, we also generated WSS with corresponding recall levels to make exact comparisons with them. Again, we used the results obtained from the flexible depth thresholds.

As shown in Table 4, we had a better WSS95 performance on average and in total than Jonnalagadda [22] (19.24% vs 13.17%,

Table 3
WSS95 (%) comparison with Cohen [10], Matwin [13], and Ji and Yen [17] systems across 15 SRs.

SR Report Topic	Cohen [10] (SVM)	Matwin [13] (FCWB/WE)	Ji and Yen [17] (Lexical Network)	Our Study (Semantic Network)
ACE Inhibitors	73.3	52.3	74.29	85.57
ADHD	52.6	62.2	67.92	83.55
Antihistamines	23.6	14.9	24.52	23.87
Atypical Antipsychotics	17	20.6	17.95	35.18
Beta Blockers	46.5	36.7	65.01	68.48
Calcium Channel Blockers	43	23.4	17.32	25.53
Estrogens	41.4	37.5	22.55	32.07
NSAIDS	67.2	52.8	77.35	64.89
Opioids	36.4	55.4	8.98	17.39
Oral Hypoglycemic	13.6	8.5	13.52	15.90
Proton Pump Inhibitors	32.8	22.9	19.65	25.81
Skeletal Muscle Relaxants	37.4	26.5	58.55	36.15
Statins	49.1	31.5	27.8	43.74
Triptans	34.6	27.4	39.64	54.10
Urinary Incontinence	43.2	29.6	20.8	44.95
Average	**40.8**	**33.5**	**37.06**	**43.81**
Total	**611.70**	**502.20**	**555.85**	**657.18**

SVM: Support Vector Machine.
FCWB/WE: Factorized Complement Naïve Bayes with Weight Engineering.

Table 4
WSS comparison between our semantic approach and other semantic approaches.

SR Report Topic	Abstract level decisions WSS95 (%)		Full-text level decisions WSS (%) at various recall levels		
	Jonnalagadda [22]	Our Study	[a]*Recall*	Khabsa [14]	Our Study
ACE Inhibitors	13.40	18.00	*98.6*	46.90	53.34
ADHD	29.63	78.03	*98.00*	44.70	81.55
Antihistamines	16.45	22.90	*75.00*	3.00	37.42
Atypical Antipsychotics	6.07	9.11	*96.00*	19.90	28.39
Beta Blockers	7.24	9.51	*94.70*	36.10	68.48
Calcium Channel Blockers	8.13	9.77	*98.00*	28.70	16.34
Estrogens	30.16	32.07	*98.30*	18.00	16.03
NSAIDS	12.98	20.87	*99.50*	40.40	52.93
Opioids	6.11	5.12	*93.30*	45.50	39.11
Oral Hypoglycemic	15.31	16.30	*97.10*	7.40	5.77
Proton Pump Inhibitors	10.80	13.73	*97.60*	28.80	21.08
Skeletal Muscle Relaxants	6.15	16.37	*82.20*	37.10	50.64
Statins	15.15	16.74	*94.10*	40.00	44.54
Triptans	13.56	13.11	*95.80*	31.20	54.10
Urinary Incontinence	6.42	7.03	*94.00*	41.10	44.95
Average	**13.17**	**19.24**	***94.15***	**31.25**	**40.98**
Total	**197.56**	**288.66**	–	**468.8**	**614.68**

[a] Various recall levels used in Khabsa [14], ranging from 75.00% to 99.50%.

and 288.66% vs 197.56%) and Khabsa [14] (40.98% vs 31.25%, and 614.68% vs 468.8%). With pairwise comparisons, we did not find a significant difference between our approach and Jonnalagadda [22] ($p = 0.072$), but found a significant difference between our approach and Khabsa [14] ($p = 0.031$). In addition, when examining each individual SR, we outperformed Jonnalagadda [22] in 13 SRs, and outperformed Khabsa [14] in 10 SRs. In summary, our UMLS ontology-based semantic approach provided a better performance in identifying relevant articles in the biomedical domain than the existing corpus-based semantic approaches.

4. Discussion

4.1. Customizable and interpretable parameters

Due to the fact that different SRs have various research scopes, we could not identify a universal parameter setting (i.e. depth threshold) that could be successfully applied to every SR [17]. A similar thought was stated in Matwin's work [13], where weight parameters (or weight multiplier, WM) need to be tunable and modified with regards to different SRs. Therefore, instead of finding the best parameter for each SR, providing flexible and customizable parameters for SR reviewers based on SR scopes and screening priority is the most useful and practical approach. A tunable depth threshold is especially helpful to meet different criteria or expectations in concept specificity. At the beginning of an article screening process, SR researchers might set a higher depth threshold (high specificity) to rapidly go to a specific scope for their best interest; they can then gradually lower the threshold (lower specificity) to find other relevant studies that may not contain as specific concepts. However, worth mentioning, not only did our flexible depth thresholds result in an outstanding performance, our approach also provided competitive results using a fixed depth threshold. In addition, it is likely that we could further improve the performance through adjustable weights for concept categorizations (α, β) and the contribution of *title* and *abstract* ($\vec{w}$). In summary, as there is no global algorithm for different SRs, the generalizability of supervised machine learning approaches can be limited. Our article network based approach, on the other hand, provides easily interpretable and adjustable parameters for SR researchers, and thus can effectively integrate prior knowledge based on SR researchers' best interests. Our approach would be a more practical solution in providing a friendly system to assist with SRs.

4.2. Ontology-based semantics versus corpus-based semantics

Corpus-based semantic approaches use distributional semantics learned from word co-occurrences in corpus context, and they rely on a large and high-quality corpus. These approaches have been used to facilitate the identification of relevant articles in SRs [14,22], and are usually implemented via word embedding using neural networks [20], a machine learning model. Word embedding has been popularly used in NLP research to map words to semantic vectors and capture extended word relations (i.e. semantic similarity). For the linear structured word vectors, vector arithmetics (e.g. addition, average, subtraction) can be used to infer phrase, sentence, and article level relationships. However, more studies are needed to demonstrate its capabilities to model biomedical knowledge more precisely. Our ontology-based semantic approach, on the other hand, uses background knowledge maintained in UMLS to derive both concept level and article level relations. Both SNOMED-CT and MeSH were used as our knowledge sources as they have been demonstrated to enable a broader coverage of concepts and knowledge. We found that our ontology-based approach provided better performance in workload savings than previous corpus-based approaches. It is likely that because the training process is needed in the corpus-based approaches, their performance highly relies on word co-occurrences and distribution. Distributional word relations learned from corpus context might contain noises, because some words from a similar context may not necessarily have direct or meaningful relations. Previous studies also compared ontology-based methods with corpus-based methods to calculate concept (or word) similarity, and found that the ontology-based intrinsic IC [26] method is highly correlated with human ratings [26,31]. This method has been confirmed to be one of the most preferred approaches. Moreover, the implementation of corpus-based semantic approaches requires a learning process with i.e. neural networks (if not using generic models which are already trained); without a learning process, the implementation of our ontology-based method will also be able to provide higher efficiency with parallel computing.

4.3. Study limitations

We have some limitations in our study. First, we were unable to compare our semantic approach with other active learning methods, i.e., Wallace [12] and Miwa [15], because they used different datasets. Second, we only used 15 DERP SRs to demonstrate our performance. Third, we only used SNOMED-CT and MeSH from UMLS Metathesaurus. Our performance relies on the scope, coverage and quality of the ontologies, which determine the knowledge that we can obtain to annotate texts and capture contributing semantics. However, SNOMED-CT and MeSH are the most widely used standards in related studies. We consider SNOMED-CT and MeSH as sufficient resources for our current task in prompting relationships among biomedical literature and assisting with SRs. Our future studies will include additional ontologies to capture a wider range of semantics.

5. Conclusion

We demonstrated using ontology-based semantics to facilitate the identification of relevant articles for SRs. We used SNOMED-CT and MeSH from UMLS to derive effective concepts and concept relations as the foundations of our approach. We created a process to develop an optimized and enriched concepts representation for each title and abstract. Within a weighted semantic space consisting of UMLS concepts, we calculated article similarities and produced a semantic article network. Instead of using distributional semantics learned from corpuses for specific topics, our method can easily apply to any topics in the biomedical domain with background knowledge from the ontologies. Besides, without relying on prior supervised training data, our established semantic article network can be applied to assist in the article screening process for any SR efficiently and with greater generalizability.

Our results showed that using SNOMED-CT and MeSH together provided better performance than using SNOMED-CT or MeSH alone; not only did our flexible depth thresholds result in the best performance, our approach also provided competitive results when using a fixed depth threshold. When comparing to other studies, we were able to achieve an average of 43.81% WSS95, which is better than other approaches using lexical features. In addition, we also confirmed that our ontology-based semantic approach provided better performance than existing corpus-based semantic approaches. We are currently deploying the semantic approach into a publically accessible web-based system, ArticleNet, aiming to assist in the article screening process for SRs and literature reviews of any size. Our future plans include further examining our approach with SRs from other research teams and exploring other ontologies.

Authorship contributions

XJ and PY designed and planned the experiments. AR provided guidance on semantic analysis and machine learning. XJ coded programs to conduct the experiments. XJ and PY analyzed and interpreted the study findings. XJ drafted the manuscripts. PY and AR provided revisions for the manuscript. All authors read and approved the final manuscript.

Conflict of interest

The authors whose names are listed immediately below certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

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